Route Correction and Variance Decomposition for Aerobic Decoupling: A 253K-Workout Study of Durability Dimensionality

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Meixner et al. (2025) proposed four conceptual constructs for exercise durability—durability, fatigability, repeatability, and resilience—yet no large-scale empirical validation exists. To examine the empirical structure of these constructs, we operationalized three of them as author-defined field indices: the Decoupling Index (DI; durability), the Fading Index (FI; fatigability), and the Resilience Index (RI; resilience). Using 253,020 workouts (675 users, FitRec dataset), we examined two questions under the condition that DI is computed as HR/Speed—the form deployed to the majority of real-world users. CRQ1 (Effective Dimensionality): The data support a two-dimensional structure. DI and the Fading Index collapsed onto a single construct ($r = -0.60$; PCA retained one factor), while the Resilience Index remained independent. Speed-based DI was predictable from route geometry alone (CV $R^2 = 0.82$; 7-fold range with zero cardiac drift); naive power estimation via Minetti cost overcorrected in the opposite direction (CV $R^2 = 0.57$). CRQ2 (Variance Structure): The dimensionality reduction is explained by the dominance of day-to-day variation. Variance decomposition revealed that residual variance accounted for 57–83% of all HR-response metrics; route-matched sensitivity analysis reduced this to 27–53%, but the core finding persists: the same person on the same route produces a different HR response every session. Convergence analysis showed that gradient sensitivity reaches adequate reliability (SB ≥ 0.80) at ≥ 6 sessions, providing a concrete aggregation protocol. Person-level signal accounts for 22–43% of variance, recoverable through multi-session aggregation. We propose gradient sensitivity (β_{gradient}) and speed sensitivity (β_{speed}) as route-corrected durability profiling metrics, and recommend integrating daily readiness markers to account for the dominant Occasion component.

aerobic decoupling | durability | cardiac drift | heart rate | gradient sensitivity | measurement artifact | field testing

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Introduction

Aerobic decoupling—the progressive rise in heart rate (HR) relative to pace or power during prolonged exercise—has been widely adopted as a field-based indicator of endurance fitness (1, 2). Platforms such as TrainingPeaks and Strava present the Decoupling Index (DI) to millions of athletes, applying Friel's $< 5\%$ threshold to indicate adequate aerobic base. This threshold has never been empirically validated in a peer-reviewed study. The most closely related large-scale study is Smyth et al. (3), who characterised internal-to-external workload decoupling in 82,303 marathon runners and demonstrated that both decoupling magnitude and onset are associated with performance. Their work establishes the ecological validity of field-based decoupling measurement at scale, but did not examine the route-geometry artifact, the dimensional structure of multiple decoupling indices, or the variance attributable to person, route, and occasion—the gaps addressed here.

On variable terrain, speed—determined primarily by gradient—is an imperfect proxy for external workload, introducing a route-geometry component into speed-based DI. What has not been established is (a) the magnitude of this component, (b) whether it dominates physiological signal in real-world data, and (c) what meaningful signal remains once it is corrected.

Meixner et al. (4) proposed four conceptual constructs for endurance durability: durability, fatigability, repeatability, and resilience. Eight commentaries followed, including Colosio et al. (5) who discussed the physiological distinction between these constructs. All discussions have remained at the conceptual level. To bridge theory and large-scale field data, this study introduces three author-defined operationalizations of Meixner et al.'s constructs: the Decoupling Index (DI; operationalizing durability), the Fading Index (FI; fatigability), and the Resilience Index (RI; resilience). The fourth construct (repeatability) was not operationalized in this study due to the absence of repeated-bout protocols in the FitRec dataset. These indices are not defined in Meixner et al. (4); they are original operationalizations designed to map the theoretical constructs to measurable GPS and HR field data. One likely reason no prior large-scale study exists is a disciplinary gap: Meixner et al.'s framework defines DI as HR/Power, while the FitRec dataset (6)—published in the recommender-systems community—contains HR and GPS but no power data. Yet the majority of real-world users compute HR/Speed via consumer GPS watches. This study does not challenge the theoretical constructs themselves; rather, it examines the empirical structure that emerges when those constructs are operationalized with the speed-based metrics that millions of users actually employ. This distinction—between theoretical construct validity and field-level operationalization—is the central concern of the present work.

The reliability of field-based HR metrics has long been a concern (7, 8). The IOC consensus (9) emphasized the external-load/internal-response distinction, but no study has quantified the variance attributable to person, route, and occasion in HR-response metrics.

Research Questions. All analyses use speed-based DI (HR/Speed), not power-based DI (HR/Power), because no power data are available in FitRec. Results apply to the majority of real-world users who compute DI via consumer GPS watches.

CRQ1: What is the empirically supported dimensionality of the durability framework under speed-based DI?

- **H1** DI and FI collapse onto a single construct, supporting a two-dimensional structure.
- **H2** All durability dimension ICCs fall below 0.50, indicating insufficient reliability for four independent dimensions.
- **H3** DI route predictability vanishes after gradient correction (GACD), confirming that the route-geometry component can be effectively removed.

CRQ2: To what extent do stable individual physiology, route properties, and transient daily state each account for the variance in HR-response metrics?

- **H4** Transient daily state (Occasion) accounts for $> 50\%$ of variance, necessitating multi-session aggregation for reliable measurement.

Three exploratory hypotheses were also examined:

- **E1** Temporal placement of descent affects cardiac drift.
- **E2** A more stable HR parameter than DI exists.
- **E3** ICC $\ll$ split-half reliability (rank-order stability $>$ absolute consistency).

Contributions. C1 Empirical evidence that a two-dimensional structure is supported over the proposed four-construct framework, with quantification of the route-geometry artifact ($R^2 = 0.82$; 7-fold range) and demonstration that naive power estimation overcorrects. **C2** First large-scale variance decomposition (Person/Route/Occasion) for HR-response metrics: the same person on the same route shows 27–53% day-to-day variation, indicating that the four-construct operationalization requires multi-session aggregation to be reliably measured. **C3** A between-person association between descent placement and cardiac drift, with modest within-person support. **C4** Convergence analysis showing that gradient sensitivity requires ≥ 6 sessions for adequate reliability.

Methods

Data and Filtering. We used the FitRec dataset (6), comprising 253,020 GPS-tracked workouts from Endomondo. Each workout record j contains irregularly time-stamped arrays of heart rate ($\mathrm{HR}_{j,i}$, optical wrist sensor, bpm), speed ($v_{j,i}$, GPS-derived, km/h in the raw data), altitude ($a_{j,i}$, barometric or GPS-derived, m), and geographic coordinates, where $i = 1, \ldots, N_j$ indexes the data points. Where the recorded speed array was missing or shorter than 10 points, speed was derived from GPS coordinates via the Haversine formula (values exceeding 150 km/h were excluded as outliers). The gender variable indicates 94.0% male, 4.8% female, 1.2% unknown; age, body mass, and training history are unavailable. Sensor make and firmware version are not recorded.

A four-step inclusion pipeline was applied: (1) HR, altitude, and speed arrays each required $N_j \geq 30$ finite data points; missing values were linearly interpolated between flanking valid points; (2) total duration $T_j = t_{N_j} - t_1 > 90$ min; (3) altitude range $\Delta a_j = \max_i(a_i) - \min_i(a_i) > 200$ m, ensuring variable terrain; (4) each temporal half (H1: $i \leq \lfloor N_j/2 \rfloor$; H2: $i > \lfloor N_j/2 \rfloor$) required > 5 points with $v_i > 0.5$ km/h (i.e., ≥ 6 per half). When speed was entirely unavailable, a heart-rate-only DI ($= \bar{\mathrm{HR}}_{\mathrm{H2}}/\bar{\mathrm{HR}}_{\mathrm{H1}}$) was computed as a fallback; FI and RI required speed.

This yielded $N = 13{,}750$ workouts from $K = 675$ users for Study 1 (cycling 61.8%, running 16.9%, mountain biking 16.7%, other 3.6%; median duration 181 min [IQR 132–265]; median altitude range 349 m [IQR 263–545]; median HR 135 bpm [IQR 125–145]). Studies 2–3 required ≥ 5 repeated workouts per user, yielding 2,343 workouts from 314 users.

Metric Definitions. Speed in the raw data is recorded in km/h (metrics 1–3). For the within-workout regression (metrics 4–6), speed was converted to m/s by the recording software; the gradient computation in the regression uses speed in m/s and timestamps in seconds.

(1) Decoupling Index (DI). Let $\mathcal{S}_h = \{i \in \mathrm{H}_h : v_i > 0.5 \text{ km/h}\}$ for $h \in \{1, 2\}$. The DI is:

$$\mathrm{DI} = \frac{\bar{\mathrm{HR}}_{\mathcal{S}_2} / \bar{v}_{\mathcal{S}_2}}{\bar{\mathrm{HR}}_{\mathcal{S}_1} / \bar{v}_{\mathcal{S}_1}}$$

where $\bar{\mathrm{HR}}_{\mathcal{S}_h} = |\mathcal{S}_h|^{-1} \sum_{i \in \mathcal{S}_h} \mathrm{HR}_i$ and analogously for $\bar{v}$. DI > 1 indicates cardiac drift.

(2) Fading Index (FI). For FI, point-to-point gradient was computed from the cumulative horizontal distance: $D_i = \sum_{l=1}^{i} v_l \cdot \Delta t_l / 3600 \times 1000$ (m), with minimum horizontal step 0.1 m, and $g_i = \mathrm{clip}(100 \times \Delta a_i / \Delta D_i,\ -50,\ +50)$ %. The gradient range was partitioned into five bins $\mathcal{B}_b$: $[-50, -10)$, $[-10, -3)$, $[-3, 3)$, $[3, 10)$, $[10, 50)$ %. Within each bin:

$$\mathrm{FI}_b = \frac{\bar{v}_{\mathcal{S}_2 \cap \mathcal{B}_b}}{\bar{v}_{\mathcal{S}_1 \cap \mathcal{B}_b}}, \quad \mathrm{FI} = \frac{1}{|\mathcal{B}^*|} \sum_{b \in \mathcal{B}^*} \mathrm{FI}_b$$

where $\mathcal{B}^* = \{b : |\mathcal{S}_1 \cap \mathcal{B}_b| > 3 \text{ and } |\mathcal{S}_2 \cap \mathcal{B}_b| > 3\}$.

(3) Resilience Index (RI). Altitude was smoothed by a trailing moving average (window $w = \max(1, \lfloor N/20 \rfloor)$, mode = "valid"). A climb was initiated when the smoothed altitude increased by > 0.5 m from one point to the next, and terminated when this condition ceased; only climbs with cumulative vertical gain > 50 m were retained. Let $C_1, \ldots, C_m$

denote the retained climbs. If $m \geq 2$:

$$\mathrm{RI} = \bar{v}_{C_m} / \bar{v}_{C_1}$$

where $\bar{v}_{C_l} = |\{i \in C_l : v_i > 0.5\}|^{-1} \sum_{i \in C_l,\, v_i > 0.5} v_i$. Workouts with $m < 2$ received no RI.

(4–6) Within-workout regression. For the regression-derived metrics, speed was in m/s and gradient was computed as:

$$g_i = \mathrm{clip}\left(100 \times \frac{a_{i+1} - a_i}{d_i},\ -50,\ +50\right),$$
$$d_i = \max(v_i \cdot \Delta t_i,\ 0.1\ \mathrm{m}), \quad \Delta t_i = \max(t_{i+1} - t_i,\ 0.1\ \mathrm{s})$$

Note that the minimum distance floor is 0.1 m (not 0.5 m as in FI), reflecting the finer resolution needed for regression. All variables were evaluated at midpoints: $\mathrm{HR}_i^* = (\mathrm{HR}_i + \mathrm{HR}_{i+1})/2$, $v_i^* = (v_i + v_{i+1})/2$, $t_i^* = (t_i + t_{i+1})/2$ (min). On the subset of valid indices ($|\mathcal{V}| \geq 20$ required), the OLS regression:

$$\mathrm{HR}_i^* = \beta_0 + \beta_{\text{grad}}\, g_i + \beta_{\text{speed}}\, v_i^* + \beta_{\text{time}}\, t_i^* + \varepsilon_i,$$
$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

where $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times 4}$ (intercept, g, v^*, t^*) yielded: (4) GACD ($\hat{\beta}_{\text{time}}$, bpm/min); (5) Gradient sensitivity ($\hat{\beta}_{\text{grad}}$, bpm/%); (6) Speed sensitivity ($\hat{\beta}_{\text{speed}}$, bpm/(m/s)). Per-workout $R^2 = 1 - \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 / \|\mathbf{y} - \bar{y}\mathbf{1}\|^2$. Although gradient ($g_i$) and speed ($v_i^*$) exhibit collinearity in field environments, variance inflation factors remained low: median VIF was 2.01 (gradient), 2.09 (speed), and 1.07 (time), with 100% of workouts below the conventional threshold of 10 and 95.7% below 5.

Estimated Power Experiment. DI was recomputed with the denominator replaced by estimated metabolic power $P_i = v_i \cdot C_r(g_i)$ (W/kg), where $C_r(g)$ is the Minetti (10) gradient cost of transport (J/kg/m), linearly interpolated from the published 23-point table ($g = -45\%$ to $+45\%$; $C_r(0\%) = 1.60$, $C_r(-3\%) = 0.40$, $C_r(+10\%) = 6.00$):

$$\mathrm{DI}_{\text{EstP}} = \frac{\bar{\mathrm{HR}}_{\mathcal{S}_2} \,/\, \bar{P}_{\mathcal{S}_2}}{\bar{\mathrm{HR}}_{\mathcal{S}_1} \,/\, \bar{P}_{\mathcal{S}_1}}$$

After outlier exclusion (1st and 99th percentiles), $N =$ 13,029 workouts remained.

Study 1: Construct Validity — H1 (DI–FI redundancy), H2 (ICC < 0.50). To test whether the proposed four dimensions are empirically distinct (H1) and individually reliable (H2), person-level medians of DI, FI, and RI were computed for each user with ≥ 5 workouts ($N =$ 13,750). Pairwise Pearson correlations were estimated via nonparametric bootstrap ($B =$ 2,000; users resampled with replacement; 95% CI from 2.5th and 97.5th percentiles). If H1 holds, DI and FI should show a strong negative correlation (FI < 1 when DI > 1) and PCA should retain only one factor.

PCA was performed on column-standardised person-level medians ($z_{ij} = (x_{ij} - \bar{x}_j)/s_j$) for users with valid values on all three metrics. Factor retention was determined by parallel analysis (11): observed eigenvalues were compared to the 95th percentile of eigenvalues from $B = 100$ column-wise random permutations of standard normal data of the same dimensions. ICC(1,1) was computed for each metric on users with ≥ 5 workouts (see Study 3); H2 predicts all ICCs < 0.50.

Study 2: Route-Geometry Correction — H3 (gradient correction), E1 (descent placement). Route prediction. Let $\mathbf{x}_j \in \mathbb{R}^{13}$ denote 13 route features (total ascent/descent, altitude range, max/min altitude, gradient mean/std, climb/descent/flat fractions, ascent-front and descent-front ratios, duration). Within each cross-validation fold f, features were z-standardised using training-fold statistics only ($\mu_p^{(f)}$, $\sigma_p^{(f)}$ computed on train$_f$; applied to both train$_f$ and test$_f$), and Ridge regression was fitted:

$$\hat{\beta}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \alpha \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}, \quad \alpha = 1.0$$

and gradient boosted trees (100 trees, max depth 3) were fitted. Generalisation was assessed by group k-fold cross-validated R^2 ($k = \min(5, K_{\text{valid}})$; all workouts of each user in the same fold):

$$R^2_{\text{CV}} = 1 - \frac{\sum_{f=1}^{k} \sum_{j \in \text{test}_f} (y_j - \hat{y}_j^{(f)})^2}{\sum_{f=1}^{k} \sum_{j \in \text{test}_f} (y_j - \bar{y})^2}$$

The hilly subset was restricted to $\Delta a_j > 500$ m. For GACD, the target was the within-person deviation $\tilde{y}_{kj} = y_{kj} - \bar{y}_{k\cdot}$ (users with ≥ 3 workouts), isolating route effects from person-level means.

Simulation. $N =$ 5,000 synthetic workouts ($N_{\text{pts}} = 60$, split at 30) were generated with cardiac drift set exactly to zero. Five route types were sampled with equal probability:

- Front-climb: $g_{\text{H1}} \sim \mathcal{N}(8, 3^2)$, $g_{\text{H2}} \sim \mathcal{N}(-5, 3^2)$
- Back-climb: reversed
- Symmetric: $g \sim \mathcal{N}(0, 5^2)$ throughout
- Valley: $g_{\text{H1}} \sim \mathcal{N}(-6, 3^2)$, $g_{\text{H2}} \sim \mathcal{N}(6, 3^2)$
- Peak: reversed

Per-workout parameters: $\mathrm{HR}_0 \sim U(100, 140)$, $\gamma_{\text{HR}} \sim U(2, 8)$ bpm/%, $v_0 \sim U(2, 5)$ m/s, $\gamma_v \sim U(0.05, 0.15)$ (m/s)/%. The data-generating process was:

$$\mathrm{HR}_i = \mathrm{HR}_0 + \gamma_{\text{HR}} \cdot g_i + \epsilon_i^{\text{HR}}, \quad \epsilon_i^{\text{HR}} \sim \mathcal{N}(0, 3^2)$$

$$v_i = \max(0.5,\ v_0 - \gamma_v \cdot g_i + \epsilon_i^v), \quad \epsilon_i^v \sim \mathcal{N}(0, 0.3^2)$$

No time-dependent term was included; any non-unity DI arises purely from gradient asymmetry between halves.

Descent-placement analysis. Define the descent-front ratio:

$$\delta = \frac{\sum_{i \in \text{H1}} |\min(\Delta a_i, 0)|}{\sum_{i=1}^{N} |\min(\Delta a_i, 0)|}$$

This was tested as a predictor of GACD via: (a) between-person standardised coefficient β^* (controlling for total descent); (b) within-person Pearson r for users with ≥ 5 varied routes ($n = 108$); (c) steepness stratification ($\sigma_g < 5$; 5–8; > 8); (d) sport-specific subgroups. All eight exploratory p-values were adjusted by Benjamini–Hochberg FDR (12).

Study 3: Variance Decomposition — H4 (Occasion dominance), E2, E3. Study 1 establishes that no dimension reaches ICC ≥ 0.50 (H2), motivating the question of *why*. Study 3 decomposes the total variance of each HR-response metric into three sources—Person, Route, and Occasion—to test whether transient day-to-day variation is the dominant source (H4). If H4 holds, it provides a mechanistic explanation for H2: moderate ICCs arise not because the metrics are poorly defined, but because within-person session-to-session fluctuation reflects genuine physiological variability (readiness, recovery, environment) that multi-session aggregation can address. Study 3 also compares metrics to identify the most reliable candidate (E2) and examines whether rank-order consistency exceeds absolute consistency (E3).

ICC(1,1). One-way random-effects ANOVA (13) for users with $n_k \geq 5$:

$$\mathrm{ICC}(1,1) = \frac{\mathrm{MS_B} - \mathrm{MS_W}}{\mathrm{MS_B} + (n_0 - 1)\,\mathrm{MS_W}}$$

$$\mathrm{MS_B} = \frac{\sum_k n_k(\bar{x}_{k\cdot} - \bar{x}_{\cdot\cdot})^2}{K-1},$$

$$\mathrm{MS_W} = \frac{\sum_k \sum_j (x_{kj} - \bar{x}_{k\cdot})^2}{N-K},$$

$$n_0 = \frac{N - \sum_k n_k^2/N}{K-1}$$

Variance partitioning. The between-person share was $\%\mathrm{Person} = \mathrm{ICC} \times 100$. Within-person deviations $\tilde{x}_{kj} = x_{kj} - \bar{x}_{k\cdot}$ were predicted from the 13 route features via Ridge regression ($\alpha = 1.0$) with group k-fold CV ($k = \min(5, K_{\text{valid}})$), yielding R^2_{route}. Because this prediction operates only on the within-person variance $(1 - \mathrm{ICC})$:

$$\%\mathrm{Route} = \max\bigl(0,\, R^2_{\text{route}}\bigr) \cdot (1 - \mathrm{ICC}) \cdot 100,$$
$$\%\mathrm{Occasion} = 100 - \%\mathrm{Person} - \%\mathrm{Route}$$

The residual %Occasion captures daily physiological state, environmental conditions, sensor noise, and unmodeled route variation. Uncertainty in the variance partitioning was estimated via user-level bootstrap ($B = 500$; 95% CI from percentiles).

Route-matched sensitivity analysis. Workout pairs were matched if:

$$\left|\frac{\Delta a_j - \Delta a_{j'}}{\Delta a_j}\right| \leq 0.20 \quad \text{and} \quad \left|\frac{A_j^+ - A_{j'}^+}{A_j^+}\right| \leq 0.20$$

where A^+ denotes total ascent ($n = 451$ pairs, 78 users). Route-matched ICC provided an upper-bound estimate of person-level stability.

Split-half reliability. For users with ≥ 10 workouts, the Pearson r between the mean of the first half and the mean of the second half of chronologically ordered workouts was corrected via the Spearman–Brown prophecy formula: $\mathrm{SB} = 2r_{\text{sh}}/(1 + r_{\text{sh}})$.

Convergence analysis. For $k = 2, 3, \ldots, 10$, users with $\geq 2k$ workouts were selected. Workouts were randomly permuted within each user, then the first k and second k workouts were used to compute split-half r and SB_k. This estimates the reliability achievable from k randomly sampled sessions, providing guidance on the minimum sessions for $\mathrm{SB} \geq 0.80$.

ICC sensitivity. ICC was recomputed with minimum-workout thresholds $n_{\min} \in \{3, 5, 8, 10, 15\}$.

Convergent validity. User-level median speed served as an external criterion; Pearson r between each metric's user-level median and median speed was computed.

Software and reproducibility. Python 3.12 (NumPy 1.26, SciPy 1.13, pandas 2.2, scikit-learn 1.5; random seed 42). Code: https://github.com/ikari12/GRAD.

Results

Results are organised by CRQ. Study 1 and Study 2 address CRQ1 (dimensionality); Study 3 addresses CRQ2 (variance structure) and shows how multi-session aggregation achieves reliable measurement.

CRQ1: A Two-Dimensional Structure Is Supported (H1, H2). DI and FI showed $r = -0.60$ (95% CI $[-0.69, -0.49]$; bootstrap $B = 2{,}000$ on person-level medians), confirming H1: DI and FI capture the same underlying gradient-response coupling, supporting a more parsimonious two-dimensional model. PCA on standardized person-level medians retained one factor ($\text{eigenvalue}_1 = 1.61$, 53.6%; $\text{eigenvalue}_2 = 1.01$, below the parallel-analysis 95th-percentile threshold; Table 1). RI was largely independent (DI–RI: $r = -0.19$; FI–RI: $r = +0.19$), confirming a two-dimensional structure: one DI–FI factor and one RI factor (Figure 1).

All ICCs fell below 0.50: DI $= 0.16$, FI $= 0.08$, RI $= 0.10$ (H2 confirmed). This motivates the question addressed by CRQ2: what is the source of this within-person variation, and can it be managed through multi-session aggregation?

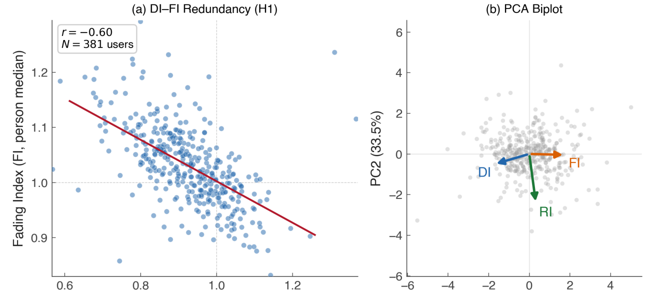

Fig. 1. (a) DI vs. FI person-level medians ($N = 381$ users with ≥ 5 workouts). The strong negative correlation ($r = -0.60$) confirms H1: both metrics respond to the same gradient asymmetry. (b) PCA biplot. DI and FI load in opposite directions on PC1 (53.6%), while RI is orthogonal on PC2, confirming a two-dimensional structure.

Table 1. PCA eigenvalues and parallel analysis on person-level medians. Loadings on PC1: DI $= -0.694$, FI $= +0.708$, RI $= +0.130$.

	Eigenvalue	% Var.	Parallel 95%	Retain?
PC1	1.613	53.6	1.028	Yes
PC2	1.007	33.5	1.006	No
PC3	0.388	12.9	0.994	No

CRQ1: Gradient Correction Successfully Isolates Cardiac Drift (H3). The DI–FI coupling observed in H1 reflects a shared response to route gradient. Study 2 confirmed this and demonstrated that regression-based correction (GACD) successfully removes the route component, isolating true cardiac drift.

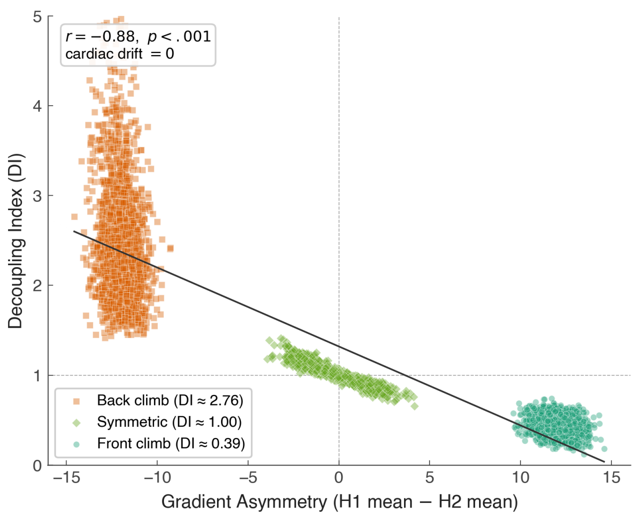

Fig. 2. DI vs. gradient asymmetry in $N = 5{,}000$ synthetic workouts with zero cardiac drift. Front-climb: DI $= 0.39$; back-climb: DI $= 2.76$; symmetric: DI $= 1.00$. $r = -0.89$.

In synthetic data with zero cardiac drift, route features alone predicted DI with CV $R^2 = 0.82$ (Figure 2). In real data, CV $R^2 = 0.58$ (all routes) and 0.60 (hilly subset). After gradient correction (GACD), route predictability was eliminated (CV $R^2 = -0.03$; H3 confirmed). This demonstrates that GACD effectively extracts the physiological signal from the route-confounded DI, and explains the DI–FI coupling in H1: both metrics respond to gradient asymmetry, which GACD removes.

A natural question is whether replacing speed with estimated metabolic power—computed as speed $\times\, C_r(g)$ where $C_r(g)$ is the Minetti (10) gradient cost function—would achieve the same correction. It does not: DI/EstPower showed a *stronger* route dependency in the opposite direction ($r = -0.76$ vs. $+0.48$ for DI/Speed; CV $R^2 = 0.57$ vs. 0.19). The overcorrection occurs because the Minetti cost function approaches zero on moderate descents ($C_r \approx 0.4$ J/kg/m at -3%), inflating the DI denominator ratio. This validates the regression-based approach (GACD) as the preferred gradient correction method.

CRQ2: Multi-Session Aggregation Achieves Reliable Measurement (H4, E2, E3). CRQ1 established the supported dimensionality and demonstrated effective gradient correction. CRQ2 identifies the variance structure and shows that multi-session aggregation overcomes within-person variability to achieve reliable measurement.

Table 2. Variance decomposition and reliability across Person / Route / Occasion axes (Study 3). Person columns: ICC = intraclass correlation coefficient ICC(1,1); Spl-h = split-half reliability (odd vs. even sessions); SB = Spearman–Brown corrected reliability. Route columns: R^2_{in} = in-sample R^2 of Ridge regression predicting within-person deviations from 13 route features; R^2_{CV} = cross-validated R^2 of the same model. Occ. % = percentage of total variance attributed to the Occasion (day-to-day) component. Val. (Speed) = Spearman correlation of person-level metric median with mean session speed (convergent validity).

	Person			**Route**		**Occ.**	**Val.**
Metric	ICC	Spl-h	SB	R^2_{in}	R^2_{CV}	%	Speed
DI	.16	.45	.63	.013	−.006	**83**	—
GACD	.22	.74	.85	.031	−.030	**78**	−.05
Grad. S.	.36	**.82**	**.90**	.128	**+.089**	58	**+.33**
Speed S.	**.43**	**.87**	**.93**	.069	−.092	57	−.13

Occasion accounted for 57–83% of variance across all metrics (H4 confirmed; Table 2). Crucially, this within-person variability is *manageable*: multi-session aggregation progressively recovers the stable between-person signal.

Route-matched ICC ($n = 451$ matched pairs) demonstrated this directly: holding route approximately constant increased Person estimates for all metrics—GACD ($0.22 \to 0.47$), gradient sensitivity ($0.36 \to 0.52$), speed sensitivity ($0.43 \to 0.73$)—showing that 16–30 percentage points of apparent Occasion variance is actually route-related and controllable.

Gradient sensitivity as the most reliable metric (E2). Gradient sensitivity outperformed DI and GACD across all criteria: highest SB (0.90), the only positive route CV R^2 ($+0.089$), and the strongest speed correlation ($r = +0.33$). The ICC–SB discrepancy (DI: 0.16 vs. 0.63; gradient sensitivity: 0.36 vs. 0.90; E3 confirmed) reveals that rank-order consistency is preserved across sessions—within-person *trends* are interpretable, and multi-session averaging stabilises absolute values.

Convergence analysis (Figures 3 and 4) quantified the practical protocol: gradient sensitivity reached SB ≥ 0.80 at $k = 6$ sessions ($\text{SB}_5 = 0.80$, $\text{SB}_{10} = 0.88$), speed sensitivity at $k = 3$ ($\text{SB}_3 = 0.81$). This provides a concrete, actionable guideline: *aggregate ≥ 6 sessions of gradient sensitivity for a reliable individual fitness estimate.*

Exploratory: Descent Placement (E1). A between-person association was observed between descent-front ratio and cardiac drift ($\beta^* = +0.29$, $p_{\text{FDR}} < .001$), moderated by steepness ($r = +0.53$ on moderate terrain vs. $+0.10$ on gentle terrain) and absent in mountain biking. Within-person evidence was modest: 66/108 users (61%) showed positive correlations, with a small mean $r = +0.089$. Seven of eight exploratory tests survived FDR correction. The between-person pattern is consistent with an eccentric-damage mechanism (14, 15), but the weak within-person support precludes causal conclusions.

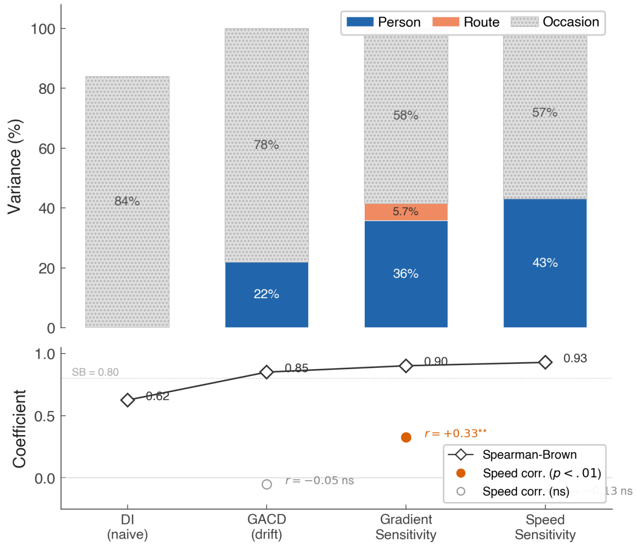

Fig. 3. Variance decomposition across Person / Route / Occasion (top) and Spearman–Brown reliability with speed correlation (bottom).

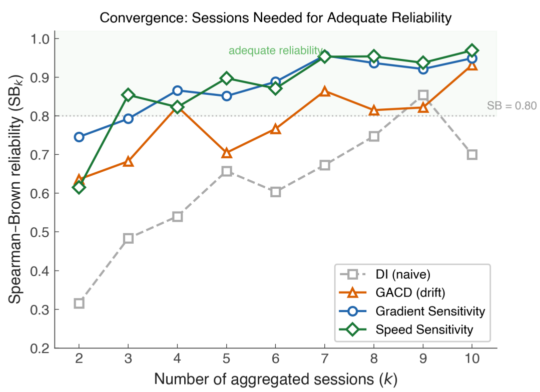

Fig. 4. Convergence curves: Spearman–Brown reliability (SB_k) as a function of the number of aggregated sessions k. The dotted line marks the SB $= 0.80$ threshold. Gradient sensitivity crosses this threshold at $k = 6$; DI requires $k \geq 9$; GACD requires $k > 10$ ($\mathrm{SB}_{10} = 0.71$).

Discussion

From Structure to Protocol: Three Converging Lines of Evidence. The three studies form a constructive progression that builds upon, rather than contradicts, the existing theoretical framework. Study 1 established that a two-dimensional structure best fits speed-based data (H1), with DI and FI capturing a single gradient-response factor and RI providing an independent dimension. This empirical convergence of DI and FI reflects a property of the specific speed-based operationalizations used here, not a theoretical prediction of the source framework: Colosio et al. (5) argued precisely that durability and fatigability are physiologically distinct constructs that should be measured separately—a distinction that our operationalization was designed to preserve but that the speed-based field metrics did not fully resolve. Study 2 showed that regression-based gradient correction (GACD) successfully isolates true cardiac drift from route geometry ($R^2 = 0.82 \to -0.03$; H3), providing the quantitative foundation that Meixner et al.'s conceptual framework requires for field deployment. Study 3 demonstrated that multi-

Table 3. Descent placement effect on cardiac drift (GACD). Exploratory p-values are FDR-adjusted.

Analysis	Statistic	Result
Total descent (amount)	r	$+.03$ ns
Descent-front (placement)	β^*	$+.29^{***}$
Within-person ($n = 108$)	Mean r	$+.089^*$
Steepness moderation		
Gentle ($\sigma < 5$)	r	$+.10^{***}$
Moderate (5–8)	r	$+.53^{***}$
Sport		
Cycling	r	$+.14^{***}$
Running	r	$+.20^*$
Mountain biking	r	$+.01$ ns

session aggregation overcomes the dominant day-to-day variability (57–83%; H4), with gradient sensitivity reaching adequate reliability at ≥ 6 sessions (SB ≥ 0.80). Friel's $< 5\%$ rule (2) represents a pioneering insight for flat or symmetric courses; GACD extends this principle to variable terrain, suggesting that Friel's aerobic-base concept may be applicable to hilly routes where naive DI is dominated by geometry, pending prospective validation. Smyth et al. (3) demonstrated in 82,303 marathon runners that decoupling magnitude and onset predict performance, confirming the practical relevance of field-based decoupling measurement; the present study extends this line of work by identifying the route-geometry component as a major confound and providing a regression-based correction (GACD) applicable beyond the road-marathon context.

Our analyses use HR/Speed, not HR/Power. This choice reflects the reality that the majority of field users compute DI from consumer GPS watches without power meters. Whether the two-dimensional structure extends to measured power-based DI remains an open question. However, the Minetti (10) estimated-power experiment provides an informative boundary: DI computed with estimated metabolic power produced a *stronger* route dependency in the opposite direction (CV $R^2 = 0.57$ vs. 0.19; $r = -0.76$ vs. $+0.48$), because the gradient cost function approaches zero on moderate descents ($C_r \approx 0.4$ J/kg/m at -3%), inflating the denominator ratio. Furthermore, the Minetti cost function was originally derived for human walking and running; its application to a dataset dominated by cycling (61.8%) introduces an inherent bioenergetic mismatch, as cycling efficiency on gradients is governed by distinct biomechanical and aerodynamic constraints (gear ratios, reduced air resistance at lower speeds on climbs). This modality mismatch further amplifies the overcorrection, underscoring that a rigid analytical cost function is difficult to adapt to mixed-sport, real-world data—whereas the data-driven regression approach (GACD) captures modality-specific submaximal relationships directly from the signal. The key contribution of GACD is therefore not contingent on the speed-vs-power distinction: it provides a general framework for isolating cardiac drift from any terrain-confounded ratio.

A Practical Protocol for Field-Based HR Monitoring. The variance decomposition (H4) provides the first quantitative evidence for the IOC consensus (9) principle that external load $\neq$ internal response. That Occasion accounts for 57–83% of variance is not merely a consequence of uncontrolled field conditions; it is precisely the point. In controlled laboratory settings, day-to-day variation is minimised by design, creating the misleading impression that a single session suffices for reliable measurement. The ecological validity of 253,020 real-world workouts reveals the true magnitude of session-to-session fluctuation that practitioners face, and provides the strongest empirical argument for the multi-session aggregation protocol we propose. The route-matched residual (27–53%) was not explained by effort intensity ($r = -0.004$, ns), likely reflecting unmeasured readiness factors such as sleep quality, hydration, temperature, and accumulated training stress (8)—each of which represents a candidate for future integration into personalised monitoring systems.

A dual-profiling framework. The three GACD regression coefficients serve conceptually distinct roles. β_{time} (GACD) is the true durability metric sensu Maunder et al. (16)—it captures the pure temporal degradation of cardiac efficiency during prolonged exercise, after removing terrain confounds. In contrast, β_{gradient} and β_{speed} are not measures of time-dependent degradation; they represent route-corrected, submaximal fitness capacity profiles that isolate an individual's baseline terrain-sensitivity from geometric artifacts. Based on the convergent evidence from Studies 1–3, we propose this dual framework: β_{time} for durability assessment, and β_{gradient} / β_{speed} for individual fitness profiling on hilly terrain. Table 4 summarises the head-to-head comparison of the profiling metrics against DI.

Table 4. Comparison of DI and the proposed GACD-derived metrics for individual profiling on hilly terrain. †Route (CV R^2) for DI is -2.30 from the Study 3 route-matched within-person prediction (Ridge regression on 13 route features, group k-fold CV); this differs from the Study 2 between-session route-predictability result (CV $R^2 = 0.58$ for raw DI) because Study 3 operates on within-person deviations after removing person-level means, a substantially harder prediction target. A negative CV R^2 indicates that route features cannot predict session-to-session DI fluctuations within a person better than the person's own mean.

Attribute	DI (cardiac drift)	Grad. sensitivity	Speed sensitivity
ICC	0.163 (Poor)	**0.358** (Fair)	**0.454** (Fair)
SB reliability	0.63	**0.90**	**0.93**
Sessions to SB ≥ 0.80	≥ 9	**6**	**3**
Route (CV R^2)	$-2.30^{\dagger}$	**+0.089**	+0.003
Speed corr. (r)	—	**+0.33**	—

The recommended protocol is: (1) apply GACD regression to remove route-geometry artifacts from each session, (2) extract gradient and speed sensitivity as the individual's durability profile, and (3) aggregate ≥ 6 sessions for stable estimates. Sport-specific baselines are required, as gradient sensitivity varies substantially across modalities (cycling $+7.25$ bpm/%, MTB $+3.83$, running $+1.49$; $F = 103.3$, $\eta^2 = 0.082$).

The ICC–SB discrepancy (E3) offers additional practical insight: rank-order consistency is preserved despite session-to-session variation. This means within-person *trends* across training blocks are interpretable, even when individual sessions vary. Coaches and athletes can track gradient sensitivity trends with confidence, provided they aggregate ≥ 6 sessions per assessment window.

Descent Placement: Suggestive but Unconfirmed (E1). Descent placement showed a between-person association with cardiac drift ($\beta^* = +0.29$), moderated by steepness and absent in mountain biking—a pattern consistent with eccentric muscle damage (14). However, within-person evidence was small ($r = 0.089$), precluding causal conclusions.

Gradient Sensitivity: A Candidate with Known Limitations (E2, C4). Gradient sensitivity showed the most balanced profile: SB reaching 0.80 at $k = 6$, route-matched ICC $= 0.52$, the only positive CV R^2 ($+0.089$), and speed correlation $r = +0.33$. The unadjusted ICC (0.36) reflects session-to-session variability addressable through aggregation; the route-matched ICC (0.52) demonstrates that controlling route selection further improves reliability. Prospective validation against VO_2max would strengthen the convergent validity evidence.

Limitations. (1) *Speed-based DI*: our DI uses HR/Speed, not HR/Power. The route artifact (C1) is specific to speed-based DI; H1, H2, H4 may differ for power-based DI. (2) Secondary data with uncontrolled sensor quality. (3) Residual–noise conflation: route-matched ICC partially addresses this, but full separation requires research-grade sensors. (4) No gold standard (VO_2max, power). (5) Demographics: 94% male; age and training history unknown. (6) Exploratory findings not pre-registered. (7) Multiple comparisons: 7/8 tests survived BH-FDR, but does not account for broader analytic flexibility. (8) Observational design. (9) Single dataset. (10) Sport bias (cycling 62%).

Conclusion

Under speed-based DI, the data support a two-dimensional durability structure rather than the proposed four dimensions (CRQ1), and gradient correction (GACD) successfully isolates true cardiac drift from route geometry. We propose gradient sensitivity and speed sensitivity—the GACD regression coefficients—as route-corrected durability profiling metrics. The key to reliable field-based measurement is multi-session aggregation (CRQ2): gradient sensitivity reaches adequate reliability (SB ≥ 0.80) at ≥ 6 sessions, providing a concrete protocol for practitioners. The Person signal (22–43%) is recoverable through aggregation; the Occasion component (27–53%) reflects day-to-day physiological variation that may carry information about readiness and recovery. Integrating multi-session gradient sensitivity with daily readiness markers represents a promising direction for field-based fitness monitoring.

Ethics Statement

This study is a secondary analysis of the publicly available FitRec dataset (6), which contains de-identified workout telemetry (heart rate, GPS coordinates, speed, altitude) voluntarily uploaded by Endomondo users. No personally identifiable information (names, dates of birth, or contact details) is present in the dataset. The original data collection was governed by Endomondo's terms of service and privacy policy at the time of upload. As a retrospective analysis of fully anonymised, publicly released data, this study did not require institutional ethics review under standard guidelines for secondary data research. No intervention was performed, and no individual participant can be identified from the analyses or results presented.

Data Availability

All analysis code and reproducibility scripts are available at https://github.com/ikari12/GRAD. The FitRec dataset is described in Ni et al. (2019) (6).

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